SANJAEY SHUNMUGA SUNDARAM

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EDUCATION

North Carolina State University, Raleigh NC — Aug 2024 - May 2026
Master of Computer Science — *GPA 3.4*

- Coursework: Software Engineering, Computer Networks, DevOps, Cloud Computing, Algorithms

Anna University, India — Aug 2019 - June 2023
Bachelor of Engineering, Computer Science and Engineering — *GPA 3.6*

PROFESSIONAL EXPERIENCE

Software Development Engineer — Aug 2023 - Feb 2024
Thermelgy (Energy ETA) — *Chennai, India*

- Led the design of ML-driven HVAC automation solutions that optimized efficiency and reduced energy costs by 10%
- Analyzed multi-site data using Python, Pandas, Excel and Tableau to identify issues and inform operational strategies
- Developed a reusable Python framework for cross-site data analysis, reducing report turnaround time by 20%
- Drove testing and validation of ESP32-based IoT gateway workflows to improve reliability and ensure data continuity
- Delivered $5,000 in cost savings by building a web app for remote IoT gateway access and resets, reducing site visits

Software Developer Intern — Jul 2021 - Oct 2021
SM Creatives — *Chennai, India*

- Developed a full-stack e-commerce platform using Vue.js, Node.js, Express.js and MongoDB
- Designed responsive frontend components and integrated backend APIs to support core shopping and user workflows
- Drove end-to-end feature development, including frontend implementation, backend logic, and database integration
- Partnered with cross-functional teams to ship high-quality features in a fast-paced environment

PROJECTS

Zero Downtime Deployment Pipeline - ZDDP

- Built a zero-downtime CI/CD pipeline for a Node.js app using GitHub Actions, Docker, Ansible, and Nginx.
- Implemented blue-green deployments with automatic rollback, test and coverage gates (≥85%), and Prometheus/Grafana monitoring.

Vision Transformer-Based Image Captioning

- Trained a CNN-RNN image captioning model achieving BLEU 0.92 and METEOR 0.90

Expense Management and Analytics App

- Developed a full-stack expense tracking app with React Native, Node.js, and MongoDB
- Led frontend design to build a responsive and intuitive user experience

Check Data Extraction (Cameo Corporate Services Ltd - Chennai)

- Built a document-processing tool using AWS Textract to extract structured bank check data into CSV/Excel, automating financial workflows

SKILLS

Languages: Python, C, C++, Java, SQL, JavaScript, HTML, CSS

Frameworks & Tools: Node.js, Express.js, Vue.js, React.js, GitHub Actions, Docker, Ansible, MongoDB, Pandas, Tableau, Excel

LEADERSHIP AND ACHIEVEMENTS

- Team Lead - ZDDP Project at NCSU (Dec 2025), IBM Applied Data Science Project (Dec 2022)
- Led a Data Analytics workshop as a Guest Speaker organized by SVCE (Oct 2023)
- Senior Executive Member - SVCE Cyber Hub (May 2023)
- Mentor - hackML'22 Hackathon (August 2022)